Notes on *Modern Quantum Mechanics* by Sakurai & Napolitano

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1 Fundamental Concepts

2 Quantum Dynamics

3 Theory of Angular Momentum

3.1 Rotation and Angular Momentum Commutation Relations

3.1.1 Finite Versus Infinitesimal Rotations

It is trivial that a 30-degree rotation about an axis followed by a 60-degree rotation about the same axis is equivalent to starting with 60- and ending with 30-degrees of rotation, thus rotations about the same axis commute. The same is not true for different axes of rotation, i.e. a 90° rotation about the x-axis followed by a 90° rotation about the y-axis is not equivalent to the reverse. The rotation of a vector $\mathbf{V}$, can be described by the orthogonal matrix $R_n(\varphi)$ such that a φ rotation about the axis n is given by

$$\mathbf{V}' = R_n(\varphi)\mathbf{V}. \tag{3.1}$$

The orthogonality of $R_n(\varphi)$ implies that $R_n^{-1} = R_n^T$ and

$$\sqrt{V_x^2 + V_y^2 + V_z^2} = \sqrt{V_x'^2 + V_y'^2 + V_z'^2}. \tag{3.2}$$

The convention used in these notes as well as the literature is that the coordinate system is static while the physical system rotates (active rotation), and positive rotation is defined as counterclockwise, i.e. the right-hand rule applies. For a rotation about the z-axis, it can be shown that

$$R_z(\varphi) = \begin{pmatrix} \cos\varphi & -\sin\varphi & 0 \\ \sin\varphi & \cos\varphi & 0 \\ 0 & 0 & 1 \end{pmatrix} \tag{3.3}$$

If we want tot look at an infinitesimal rotation $\varphi = \varepsilon$, where $\varepsilon^3 = 0$, we can perform a Taylor expansion of the rotation matrix yielding

$$R_z(\varepsilon) = \begin{pmatrix} 1 - \frac{\varepsilon^2}{2} & -\varepsilon & 0 \\ \varepsilon & 1 - \frac{\varepsilon^2}{2} & 0 \\ 0 & 0 & 1 \end{pmatrix} \tag{3.4}$$

$$R_x(\varepsilon) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 - \frac{\varepsilon^2}{2} & -\varepsilon \\ 0 & \varepsilon & 1 - \frac{\varepsilon^2}{2} \end{pmatrix} \tag{3.5}$$

$$R_y(\varepsilon) = \begin{pmatrix} 1 - \frac{\varepsilon^2}{2} & 0 & \varepsilon \\ 0 & 1 & 0 \\ -\varepsilon & 0 & 1 - \frac{\varepsilon^2}{2} \end{pmatrix}. \tag{3.6}$$

By performing two successive infinitesimal rotation about the x- and y-axis, we find that

$$R_x(\varepsilon)R_y(\varepsilon) = \begin{pmatrix} 1-\frac{\varepsilon^2}{2} & 0 & \varepsilon \\ \varepsilon^2 & 1-\frac{\varepsilon^2}{2} & -\varepsilon \\ -\varepsilon & \varepsilon & 1-\frac{\varepsilon^2}{2} \end{pmatrix} \tag{3.7}$$

$$R_y(\varepsilon)R_x(\varepsilon) = \begin{pmatrix} 1-\frac{\varepsilon^2}{2} & 0 & \varepsilon \\ -\varepsilon^2 & 1-\frac{\varepsilon^2}{2} & -\varepsilon \\ -\varepsilon & \varepsilon & 1-\frac{\varepsilon^2}{2} \end{pmatrix}, \tag{3.8}$$

which shows that the commutation relation

$$[R_x(\varepsilon), R_y(\varepsilon)] = \begin{pmatrix} 0 & -\varepsilon^2 & 0 \\ \varepsilon^2 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} = R_z(\varepsilon^2) - 1 = R_z(\varepsilon^2) - R_n(0), \tag{3.9}$$

notably we find that the rotation commute if we ignore terms of order ε^2 and higher.

3.1.2 Infinitesimal Rotations in Quantum Mechanics

The previous subsection did not involve quantum mechanical concepts, which we will introduce now. Since a rotation is active, we change the system, and thus the state ket after a rotation should be different from the original one. For a rotation characterized by the matrix R, we introduce the operator $\mathscr{D}(R)$ such that

$$\alpha_R = \mathscr{D}(R)\alpha. \tag{3.10}$$

Note that the matrix representation of $\mathscr{D}(R)$ is not the same as R, but is instead dependent on the dimensionality of the ket-space of the system, i.e. for a spin $\frac{1}{2}$ system we have a 2×2 matrix while for a spin 1 system we have a 3×3 matrix representation of $\mathscr{D}(R)$.

To construct this operator we start in the same way as for the construction of a translation and time evolution operator by looking at an infinitesimal rotation

$$U_\varepsilon = 1 - iG\varepsilon, \tag{3.11}$$

where G is a generator of the rotation. From classical mechanics we know that the generator of rotation is angular momentum, and we thus have

$$G \to \frac{\mathbf{J}\cdot\hat{\mathbf{n}}}{\hbar}, \quad \varepsilon \to d\varphi, \tag{3.12}$$

where $\hat{\mathbf{n}}$ is the axis of rotation, and we rotate around this axis by an angle $d\varphi$, and we note that the angular momentum operator $\mathbf{J}$ is defined by this expression. Taking $\mathbf{J}$ to be Hermitian we guarantee $\mathscr{D}(R)$ to be unitary. We can thus write the operator for an infinitesimal rotation as

$$\mathscr{D}(\hat{\mathbf{n}}, d\varphi) = 1 - \frac{i}{\hbar}\mathbf{J}\cdot\hat{\mathbf{n}}d\varphi. \tag{3.13}$$

Note that this formalism is applicable for any type of angular momentum, and we therefore do not take $\mathbf{J}$ to be defined by $\mathbf{x}\times\mathbf{p}$ as in classical mechanics, but instead let J be defined such that the above expression is true for an infinitesimal rotation.

3.1.3 Finite Rotations in Quantum Mechanics

If we want to consider a finite rotation we perform N successive infinitesimal rotations of the form in Eq. (3.13). For a rotation about the z-axis we have

$$\mathscr{D}(\hat{\mathbf{z}}, \varphi) = \lim_{N\to\infty}\left[1 - \frac{i}{\hbar}J_z\frac{\varphi}{N}\right]^N \tag{3.14}$$

$$= 1 - \frac{iJ_z\varphi}{\hbar} + \frac{J_z^2\varphi^2}{2\hbar^2} + \cdots \tag{3.15}$$

$$= \exp\left(-\frac{iJ_z\varphi}{\hbar}\right). \tag{3.16}$$

We postulate that $\mathscr{D}$ has the same group properties as R, which are

$$\text{Identity:} \quad R\cdot 1 = R \implies \mathscr{D}(R)\cdot 1 = \mathscr{D}(R) \tag{3.17a}$$

$$\text{Closure:} \quad R_1R_2 = R_3 \implies \mathscr{D}(R_1)\mathscr{D}(R_2) = \mathscr{D}(R_3) \tag{3.17b}$$

$$\text{Inverses:} \quad RR^{-1} = 1 \implies \mathscr{D}(R)\mathscr{D}^{-1}(R) = 1 \tag{3.17c}$$

$$R^{-1}R = 1 \implies \mathscr{D}^{-1}(R)\mathscr{D}(R) = 1 \tag{3.17d}$$

$$\text{Associativity:} \quad R_1(R_2R_3) = (R_1R_2)R_3 \implies \mathscr{D}(R_1)[\mathscr{D}(R_2)\mathscr{D}(R_3)] = [\mathscr{D}_1\mathscr{D}_2]\mathscr{D}_3. \tag{3.17e}$$

3.1.4 Commutation Relations for Angular Momentum

The fundamental commutation relation for angular momentum is

$$[J_i, J_j] = i\hbar\varepsilon_{ijk}J_k, \tag{3.18}$$

where ε_{ijk} is the Levi-Civita symbol. When generators of infinitesimal transformations (like rotation) do not commute we call the corresponding group non-Abelian. Rotation is thus non-Abelian while translation is Abelian since $[p_i, p_j] = 0$. It is important to emphasize that to obtain the commutation relations for angular momentum we use that J_k is the rotation about the k-axis, and that rotation about different axes do not commute. Simply put, The commutation relation Eq. (3.18) summarizes all basic properties of rotations in three dimensions.

To see how the system rotates we can look at the expectation value of an operator in a state α and its rotated version $\alpha_R = \mathscr{D}_z(\varphi)\alpha$, which gives

$$\langle J_x\rangle = {}_R\alpha J_x\alpha_R = \alpha\mathscr{D}_z^\dagger(\varphi)J_x\mathscr{D}_z(\varphi)\alpha, \tag{3.19}$$

we thus need to compute

$$\mathscr{D}_z^\dagger(\varphi)J_x\mathscr{D}_z(\varphi) = \exp\left(\frac{iJ_z\varphi}{\hbar}\right)J_x\exp\left(-\frac{iJ_z\varphi}{\hbar}\right) = J_x\cos\varphi - J_y\sin\varphi, \tag{3.20}$$

and we thus see that we change the expectation value of J_x by rotating the system.

3.2 Spin $\frac{1}{2}$ Systems and Finite Rotations

Experimental evidence suggests that nature realizes the lowest possible dimensional realization of angular momentum, which for a spin $\frac{1}{2}$ system is $N = 2$. We can write the spin

angular momentum operators as

$$S_x = \frac{\hbar}{2}(|\uparrow\rangle\langle\downarrow| + |\downarrow\rangle\langle\uparrow|) = \frac{\hbar}{2}\sigma_x \tag{3.21a}$$

$$S_y = \frac{i\hbar}{2}(-|\uparrow\rangle\langle\downarrow| + |\downarrow\rangle\langle\uparrow|) = \frac{i\hbar}{2}\sigma_y \tag{3.21b}$$

$$S_z = \frac{\hbar}{2}(|\uparrow\rangle\langle\uparrow| - |\downarrow\rangle\langle\downarrow|) = \frac{\hbar}{2}\sigma_z. \tag{3.21c}$$

As shown in the previous subsection the behaviour of a rotation on the expectation value is given by

$$\langle S_x\rangle_R = \langle S_x\rangle\cos\varphi - \langle S_y\rangle\sin\varphi \tag{3.22a}$$

$$\langle S_y\rangle_R = \langle S_y\rangle\sin\varphi + \langle S_x\rangle\cos\varphi \tag{3.22b}$$

$$\langle S_z\rangle_R = \langle S_z\rangle, \tag{3.22c}$$

where we do not see a change in $\langle S_z\rangle$ since S_z commutes with the rotation operator $\mathscr{D}_z(\varphi)$. We see that the behaviour is that of a classical vector rotating in the xy-plane, and is general of any J_k and not just for spin.

Examining the rotation operator acting on a general ket

$$\alpha = \langle\uparrow|\alpha\rangle\uparrow + \langle\downarrow|\alpha\rangle\downarrow, \tag{3.23}$$

we find

$$\mathscr{D}_z(\varphi)\alpha = e^{-i\varphi/2}\langle\uparrow|\alpha\rangle\uparrow + e^{i\varphi/2}\langle\downarrow|\alpha\rangle\downarrow, \tag{3.24}$$

which has the interesting consequence (due to the half-angle) that a full 2π-rotation does not return the system to its original state,

$$\mathscr{D}_z(2\pi)\alpha = -\alpha, \tag{3.25}$$

such that we need a 4π-rotation. For the calculation of an expectation value, both the bra and ket are rotated the same amount and thus the mins signs cancels. The question is therefore if this is possible to observe experimentally?

3.2.1 Spin Precession Revisited

For a magnetic field $\mathbf{B} = (0, 0, B_z)^T$ and the spin interaction Hamiltonian

$$H = -\frac{e}{m_e c}\mathbf{S}\cdot\mathbf{B} = \omega S_z, \quad \omega = \frac{|e|B_z}{m_e c}, \tag{3.26}$$

where we recall that $e < 0$ for the electron. The time evolution operator for this Hamiltonian is given by

$$\mathscr{U}(t, 0) = \exp\left(-\frac{iHt}{\hbar}\right) = \exp\left(-\frac{iS_z\omega t}{\hbar}\right), \tag{3.27}$$

which is the rotation operator $\mathscr{D}_z(\omega t)$, and it is clear why this Hamiltonian causes spin precession. By using the Eqs. (3.22) and (3.24) we find that the period for the precession is $\tau_{\text{precession}} = 2\pi/\omega$ and for the state ket $\tau_{\text{state ket}} = 4\pi/\omega$.

3.3 SO(3), SU(2), and Euler Rotations

3.3.1 Orthogonal Group

The simplest way to characterize a rotation is by defined the axis of rotation ($\hat{\mathbf{n}}$) and the angle rotated (φ). This requires three numbers, the azimuthal and polar angle of $\hat{\mathbf{n}}$ as well as the angle rotated φ. This is however not a good way to characterize the rotations when we want to study the group properties. Instead, we characterize the rotation with the rotation matrix R. Using the orthogonality condition $RR^T = R^T R = 1$, we find 6 independent equations which reduce the numbers needed to describe the matrix to $9 - 6 = 3$, which is the same number as the simple way.

We say that the set of all multiplication operations with orthogonal matrices form a group, which mean that the following conditions are satisfied

1. The product of any two orthogonal matrices is another orthogonal matrix, which is satisfied because
$$(R_1R_2)(R_1R_2)^T = R_1R_2R_2^TR_1^T = 1. \tag{3.28}$$

2. The associative law
$$R_1(R_2R_3) = (R_1R_2)R_3 \tag{3.29}$$
holds.

3. The identity matrix, physically corresponding to no rotation, is a member of the class of all orthogonal matrices.

4. The inverse matrix, physically corresponding to a rotation in the reverse direction, is also a member.

This group is called Special Orthogonal group of dimension 3, or SO(3). The name special is due to only including rotation operations, and not for example the inverse operation.

3.3.2 Unitary Unimodular Group

We can characterize a rotation by a 2×2 matrix acting on a spinor

$$\chi = \begin{pmatrix} \langle\uparrow|\alpha\rangle \\ \langle\downarrow|\alpha\rangle \end{pmatrix} = \begin{pmatrix} c_+ \\ c_- \end{pmatrix} = c_+\chi_+ + c_-\chi_- \tag{3.30}$$

with $|c_+|^2 + |c_-|^2 = 1$. The most general unitary unimodular matrix can be written as

$$U(a,b) = \begin{pmatrix} a & b \\ -b^* & a^* \end{pmatrix} \tag{3.31}$$

where $a, b \in \mathbb{C}$ and satisfy the unimodular condition

$$|a|^2 + |b|^2 = 1, \tag{3.32}$$

and the matrix fulfils the unitarity condition

$$U^\dagger(a,b)U(a,b) = 1. \tag{3.33}$$

This matrix can be used to describe a rotation of a spin $\frac{1}{2}$ system.

The complex parameters are called the Cayley-Klein parameters, and is not quantum mechanical but was used in classical mechanics of rotation before the birth of quantum mechanics.

Other properties of unitary unimodular matrices are

$$U(a_1, b_1)U(a_2, b_2) = U(a_1a_2 - b_1b_2^*, a_1b_2 + a_2^*b_1) \tag{3.34}$$

fulfilling the unimodular condition, and the inverse

$$U^-1(a, b) = U(a^*, -b). \tag{3.35}$$

This group is called Special Unitary group of dimensionality 2, or SU(2). A general Unitary matrix can be written as

$$U = e^{i\gamma}\begin{pmatrix} a & b \\ -b^* & a^* \end{pmatrix}, \quad |a|^2 + |b|^2 = 1, \quad \gamma^* = \gamma. \tag{3.36}$$

Even though both SO(3) and SU(2) can be used to characterize a rotation, they are not isomorphic, since SO(3) have a period of 2π while SU(2) have a period of 4π with a 2π period being equal to minus the identity matrix. One can also say that both $U(a, b)$ and $U(-a, -b)$ map to the same rotation in SO(3).

3.3.3 Euler Rotations

The Euler rotations is a way to characterize rotation by three angles called Euler angles, which is commonly used in rigid-body classical mechanics. The idea of Euler rotations is to rotate about the z-axis, then the y-axis, and once again the z-axis. Conventionally, in classical mechanics you change the y-axis for the x-axis, as well as rotate about the body axes instead of the fixed axes. This total rotation may be described as

$$R(\alpha, \beta, \gamma) = R_z(\alpha)R_y(\beta)R_z(\gamma). \tag{3.37}$$

There is a correspondence to the above relation but for rotation operators, which is

$$\mathscr{D}(\alpha, \beta, \gamma) = \mathscr{D}_z(\alpha)\mathscr{D}_y(\beta)\mathscr{D}_z(\gamma) \tag{3.38}$$

acting on a ket-space. For a spin $\frac{1}{2}$ system we get

$$\mathscr{D}(\alpha, \beta, \gamma) = \exp\left(-\frac{i\sigma_3\alpha}{2}\right)\exp\left(-\frac{i\sigma_2\beta}{2}\right)\exp\left(-\frac{i\sigma_3\gamma}{2}\right) \tag{3.39}$$

$$= \begin{pmatrix} e^{-i(\alpha+\gamma)/2}\cos(\beta/2) & -e^{-i(\alpha-\gamma)/2}\sin(\beta/2) \\ e^{i(\alpha-\gamma)/2}\sin(\beta/2) & e^{i(\alpha+\gamma)/2}\cos(\beta/2) \end{pmatrix}, \tag{3.40}$$

which is a SU(2) matrix. This matrix is called the $J = 1/2$ irreducible representation of rotation operator $\mathscr{D}(\alpha, \beta, \gamma)$, and we have the matrix elements

$$\mathscr{D}^{1/2}_{m'm}(\alpha, \beta, \gamma) = \left\langle j = 1/2, m' \middle| \exp\left(-\frac{iJ_z\alpha}{\hbar}\right)\exp\left(-\frac{iJ_y\beta}{\hbar}\right)\exp\left(-\frac{iJ_z\gamma}{\hbar}\right) \middle| j = 1/2, m \right\rangle \tag{3.41}$$

3.4 Density Operators and Pure Versus Mixed Ensembles

3.4.1 Polarized Versus Unpolarized Beams

Up to this point the formalism deals with an ensemble of identically prepared kets $|\alpha\rangle$. We have however not dealt with situations where, say 60 % of the ensemble is prepared in $|\alpha\rangle$ and 40 % in $|\beta\rangle$, with $\alpha \neq \beta$. To solve this we introduce the concept of fractional population with probability weights, where we have some number $w_\uparrow$ being the probability of the system being prepared in the state $|\uparrow\rangle$ as well as $w_\downarrow = 1 - w_\uparrow$. Note that these are real numbers that lack phase information and are thus inherently different from a quantum mechanical superposition and instead is related to classical probability theory.

If an ensemble is completely randomized, it is called an unpolarized beam, which in the spin $\frac{1}{2}$ example would be $w_\uparrow = w_\downarrow = 0.5$. If we have a pure ensemble it would mean that we only have one spin orientation and thus $w_\uparrow = 1$ or $w_\downarrow = 1$, and we call this for a polarized beam. We could also imagine having something in between these two extremes, which is what we call a mixed ensemble or a partially polarized beam. Note that the different kets need not be orthogonal.

3.4.2 Ensemble Averages and Density Operator

To solve the problem of not just dealing with pure ensembles we introduce the density matrix formalism, which is a general formalism valid for any type of quantum mechanical system, but we will keep the examples in the spin $\frac{1}{2}$ system. We will consider an ensemble where a w_1 fraction of the system is in state $\left|\alpha^{(1)}\right\rangle$, a w_1 fraction is in state $\left|\alpha^{(2)}\right\rangle$, and so on, with the normalization condition

$$\sum_i w_i = 1, \tag{3.42}$$

and there is no requirement on orthogonality for the kets. There is also no limitation of the number of terms i regarding the dimensionality of the system. Given an operator A with eigenkets $|a'\rangle$, the ensemble average is

$$[A] = \sum_i w_i \langle\alpha^{(i)}|A|\alpha^{(i)}\rangle = \sum_i \sum_{a'} w_i \left|\langle a'|\alpha^{(i)}\rangle\right|^2 a'. \tag{3.43}$$

Note that there are two types of probability concepts here. First we have the quantum mechanical probability of the expectation value of A with respect to $\left|\alpha^{(i)}\right\rangle$ and then a classical probability of the system being in that state. Taking this and expanding into a general basis $\{|b'\rangle\}$ we get

$$[A] = \sum_i w_i \sum_{b'b''} \langle\alpha^{(i)}|b'\rangle\langle b'|A|b''\rangle\langle b''|\alpha^{(i)}\rangle \tag{3.44}$$

$$= \sum_{b'b''} \left(\sum_i w_i \langle b''|\alpha^{(i)}\rangle\langle\alpha^{(i)}|b'\rangle \right) \langle b'|A|b''\rangle, \tag{3.45}$$

where the number of terms in the b' and b'' sums are equal to the dimensionality of the ket space. We see that we have a prefactor in front of the factor dependent on A. Thus, we introduce the density operator

$$\rho = \sum_i w_i |\alpha^{(i)}\rangle\langle\alpha^{(i)}|. \tag{3.46}$$

We obtain the matrix elements of this using

$$\langle b''|\rho|b'\rangle = \sum_i w_i \langle b''|\alpha^{(i)}\rangle\langle\alpha^{(i)}|b'\rangle \tag{3.47}$$

with which we can return to the equation of the ensemble average and obtain

$$[A] = \sum_{b'b''} \langle b''|\rho|b'\rangle\langle b'|A|b''\rangle = \sum_{b''} \langle b''|\rho A|b''\rangle = \mathrm{Tr}\{\rho A\}. \tag{3.48}$$

Two important properties of the density matrix is that it is Hermitian and that

$$\mathrm{Tr}\{\rho\} = 1. \tag{3.49}$$

For a spin $\frac{1}{2}$ system we have dimensionality $N = 2$ and thus the density matrix is a 2×2 matrix. The imposed conditions mean that three independent real parameters are needed to characterize the density matrix. For this system the needed numbers are $[S_x]$, $[S_y]$, and $[S_z]$. Note that a mixed ensemble may be decomposed into pure ensembles in multiple ways, and there is no uniqueness. For a pure ensemble, meaning that there exist a $w_i = 1$ have the property

$$\rho^2 = \rho, \tag{3.50}$$

and

$$\mathrm{Tr}\{\rho^2\} = 1, \tag{3.51}$$

and the eigenvalues are either 0 or 1. For a mixed ensemble we always have

$$0 < \mathrm{Tr}\{\rho^2\} < 1. \tag{3.52}$$

3.4.3 Time Evolution of Ensembles

Let us now consider the time evolution of the density operator. Barring external interaction, we cannot change the relative weights w_i, but the kets $|\alpha^{(i)}\rangle$ will evolve in time according to the Schrödinger equation. We find from the Schrödinger equation that

$$i\hbar\partial_t\rho = -[\rho, H], \tag{3.53}$$

which looks like the Heisenberg equation of motion but with a sign flip. Important to note however, is that the Heisenberg equation of motion deals with the time evolution of operators, while the equation of motion for the density operator is derived from the Schrödinger picture and is therefore in the Schrödinger picture and not in the Heisenberg picture, even though it is an operator. We also note that this is a quantum mechanical analogue of Liouville's theorem in classical mechanics

$$\partial_t\rho_{\text{classical}} = -[\rho_{\text{classical}}, H]_{\text{classical}} \tag{3.54}$$

3.4.4 Continuum Generalizations

It is possible to generalize this formulation for ket spaces spanned by continuous variables. Consider the ket space spanned by the position eigenkets $|\mathbf{x}'\rangle$. We find that the ensemble average is given by

$$[A] = \int d^3x' \int d^3x'' \langle\mathbf{x}''|\rho|\mathbf{x}'\rangle\langle\mathbf{x}'|A|\mathbf{x}''\rangle, \tag{3.55}$$

and we find that the density matrix is described with

$$\langle \mathbf{x}''|\rho|\mathbf{x}'\rangle = \langle \mathbf{x}''|\left(\sum_i w_i \big|\alpha^{(i)}\big\rangle\big\langle\alpha^{(i)}\big|\right)|\mathbf{x}'\rangle = \sum_i w_i \psi_i(\mathbf{x}'')\psi_i^*(\mathbf{x}') \tag{3.56}$$

where the wave function appears. Note that the diagonal elements is the weighted sum of probability densities.

3.4.5 Quantum Statistical Mechanics

We will go through some properties of completely random and completely pure density matrices. Firstly, a completely random N-dimensional density matrix has the form

$$\rho = \frac{1}{N}\begin{pmatrix} 1 & 0 & 0 & \cdots & 0 \\ 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & 1 \end{pmatrix}, \tag{3.57}$$

while a pure density matrix has the form

$$\rho = \begin{pmatrix} 0 & \dots & & & \dots & 0 \\ \vdots & \ddots & & & & \vdots \\ & & 0 & & & \\ & & & 1 & & \\ & & & & 0 & \\ \vdots & & & & & \ddots \\ 0 & \dots & & & \dots & 0 \end{pmatrix}. \tag{3.58}$$

These two matrices are very different, and we introduce the quantity

$$\sigma = -\mathrm{Tr}\{\rho \ln \rho\} \tag{3.59}$$

that characterizes this difference. Writing this quantity in the diagonal basis yields

$$\sigma = -\sum_k \rho_{kk}^{\text{diagonal}} \ln \rho_{kk}^{\text{diagonal}}, \tag{3.60}$$

and it follows from this that σ must be positive semidefinite, since $0 < \rho_k k < 1$ for all k. For a completely random density matrix we find

$$\sigma = -\sum_{k=1}^{N} \frac{1}{N} \ln\left(\frac{1}{N}\right) = \ln N, \tag{3.61}$$

while for a pure ensemble we find

$$\sigma = 0. \tag{3.62}$$

We say that this is a measure of the disorder for a system, and that $\ln N$ is the maximum value this disorder can take for a density matrix under the normalization constraint. We find that σ is related to the entropy per constituent member, S, by the relation

$$S = k_B \sigma, \tag{3.63}$$

where k_b is the Boltzmann constant. Furthermore, we take this to be the definition of entropy in quantum statistical mechanics.

We make the assumption that nature wants to maximize this entropy, under the constraint that $[H]$ has a certain value. With thermal equilibrium we have

$$\partial_t \rho = 0, \tag{3.64}$$

which means that ρ and H commute, via the von Neumann equation of motion, and thus may both be diagonalized. That is, we can write ρ in the basis of the energy eigenkets. Using calculus of variations we find that

$$\rho_{kk} = \frac{\exp(-\beta E_k)}{\sum_l^N \exp(-\beta E_l)}, \tag{3.65}$$

giving the fractional population of an energy eigenstate, where

$$\beta = \frac{1}{k_B T}. \tag{3.66}$$

This is known in statistical mechanics as the canonical ensemble. A completely random ensemble is the limit of $\beta \to 0$, or the high temperature limit. We also note that the denominator is the partition function, Z, from statistical mechanics. Thus, if we have ρ in the energy eigenbasis we can write

$$\rho = \frac{e^{-\beta H}}{Z}. \tag{3.67}$$

This can then be used to calculate the ensemble average

$$[A] = \frac{\mathrm{Tr}\{e^{-\beta H} A\}}{Z} = \frac{\sum_k^N \langle A \rangle_k \exp(-\beta E_k)}{\sum_k^N \exp(-\beta E_k)}. \tag{3.68}$$

Calculating the internal energy per constituent gives

$$U = [H] = \frac{\sum_k^N E_k \exp(-\beta E_k)}{\sum_k^N \exp(-\beta E_k)} = -\partial_\beta \ln Z, \tag{3.69}$$

which is a well known formula in statistical mechanics. In the low temperature limit of $\beta \to \infty$ we obtain the pure ensemble, with only the ground state populated.

3.6 Orbital Angular Momentum

3.6.1 Orbital Angular Momentum as Rotation Generator

In the previous section we defined angular momentum as the generator of infinitesimal rotation. If the system lacks spin we can also define the angular momentum in terms of the orbital angular momentum

$$\mathbf{L} = \mathbf{x} \times \mathbf{p}, \tag{3.70}$$

which we will discuss in this section.

This definition of orbital angular momentum stasifies the known angular momentum commutation relations

$$[L_i, L_j] = i\varepsilon_{ijk}\hbar L_k, \tag{3.71}$$

which can be proven from the commutation relation between $\mathbf{x}$ and $\mathbf{p}$. We have $L_z = xp_y - yp_x$, and we study if this can be used to think about this as a generator of rotation by using the fact that momentum is the generator of translation. Thus,

$$\left[1 - i\frac{\delta\varphi}{\hbar}L_z\right]|\mathbf{x}'\rangle = \left[1 - i\frac{p_y}{\hbar}\delta\varphi x' + i\frac{p_x}{\hbar}\delta\varphi y'\right]|\mathbf{x}'\rangle \tag{3.72}$$

$$= |x' - y'\delta\varphi, y' + x'\delta\varphi, z'\rangle, \tag{3.73}$$

which is an infinitesimal rotation around the z-axis. If the momentum operator is a generator of translation, we have shown that the orbital angular momentum operator is a generator of rotation. Switching to spherical coordinates, and suppose we have a wave function $\langle\mathbf{x}|\alpha\rangle$ for an arbitrary system ket $|\alpha\rangle$ without spin, we can find the rotated wave function

$$\langle\mathbf{x}'|\left[1 - i\frac{\delta\varphi}{\hbar}L_z\right]|\alpha\rangle = \langle r, \vartheta, \varphi - \delta\varphi|\alpha\rangle \tag{3.74}$$

$$= \langle\mathbf{x}'|\alpha\rangle - \delta\varphi\partial_\varphi\langle\mathbf{x}'|\alpha\rangle, \tag{3.75}$$

from which we can identify

$$\langle\mathbf{x}'|L_z|\alpha\rangle = -i\hbar\partial_\varphi\langle\mathbf{x}'|\alpha\rangle. \tag{3.76}$$

Considering rotation about the x- and y-axes we find

$$\langle\mathbf{x}'|L_x|\alpha\rangle = -i\hbar(-\sin\varphi\partial_\vartheta - \cot\vartheta\cos\varphi\partial_\varphi)\langle\mathbf{x}'|\alpha\rangle \tag{3.77}$$

$$\langle\mathbf{x}'|L_y|\alpha\rangle = -i\hbar(\cos\varphi\partial\vartheta - \cot\vartheta\sin\varphi\partial_\varphi)\langle\mathbf{x}'|\alpha\rangle. \tag{3.78}$$

We can define the ladder operators $L_\pm$ using these, and we find

$$\langle\mathbf{x}'|L_\pm|\alpha\rangle = -i\hbar e^{\pm i\varphi}(\pm i\partial_\vartheta - \cot\vartheta\partial_\varphi)\langle\mathbf{x}'|\alpha\rangle. \tag{3.79}$$

Using

$$\mathbf{L}^2 = L_z^2 + \frac{1}{2}(L_+L_- + L_-L_+), \tag{3.80}$$

we also find

$$\langle\mathbf{x}'|\mathbf{L}^2|\alpha\rangle = -\hbar^2\left[\frac{1}{\sin^2\vartheta}\partial_\varphi^2 + \frac{1}{\sin\vartheta}\partial_\vartheta(\sin\vartheta\partial_\vartheta)\right]\langle\mathbf{x}'|\alpha\rangle, \tag{3.81}$$

which, apart from the factor $1/r^2$, is the angular part of the Laplacian $\nabla^2 = \Delta$ in spherical coordinates. It is possible to show this from the kinetic energy operator as well. We start by writing the operator identity

$$\mathbf{L}^2 = \mathbf{x}^2\mathbf{p}^2 - (\mathbf{x}\cdot\mathbf{p})^2 + i\hbar\mathbf{x}\cdot\mathbf{p}, \tag{3.82}$$

and we also have

$$\langle\mathbf{x}'|\mathbf{x}\cdot\mathbf{p}|\alpha\rangle = -i\hbar r\partial_r\langle\mathbf{x}'|\alpha\rangle \tag{3.83}$$

$$\langle\mathbf{x}'|(\mathbf{x}\cdot\mathbf{p})^2|\alpha\rangle = -\hbar^2(r^2\partial_r^2\langle\mathbf{x}'|\alpha\rangle + r\partial_r\langle\mathbf{x}'|\alpha\rangle). \tag{3.84}$$

Using these we find

$$\langle\mathbf{x}'|\mathbf{L}^2|\alpha\rangle = r^2\langle\mathbf{x}'|\mathbf{p}^2|\alpha\rangle + \hbar(r^2\partial_r^2\langle\mathbf{x}'|\alpha\rangle + 2r\partial_r\langle\mathbf{x}'|\alpha\rangle). \tag{3.85}$$

Rewriting such that we get the kinetic energy operator $\mathbf{p}^2/2m$, we get

$$\frac{1}{2m}\langle \mathbf{x}'|\mathbf{p}^2|\alpha\rangle = -\frac{\hbar^2}{2m}\left(\partial_r^2\langle \mathbf{x}'|\alpha\rangle + \frac{2}{r}\partial_r\langle \mathbf{x}'|\alpha\rangle - \frac{1}{\hbar^2 r^2}\langle \mathbf{x}'|\mathbf{L}^2|\alpha\rangle\right), \tag{3.86}$$

and since we have

$$\frac{1}{2m}\langle \mathbf{x}'|\mathbf{p}^2|\alpha\rangle = -\frac{\hbar^2}{2m}\nabla'^2\langle \mathbf{x}'|\alpha\rangle, \tag{3.87}$$

we can identify $\langle \mathbf{x}'|\mathbf{L}^2|\alpha\rangle$ as the angular part of the Laplacian.

3.6.2 Spherical Harmonics

We consider a spinless particle in a spherical symmetrical potential. The wave function is separable in spherical coordinates, and we can write the energy eigenfunctions as

$$\langle \mathbf{x}'|n,l,m\rangle = R_{nl}(r)Y_l^m(\vartheta,\varphi), \tag{3.88}$$

which is a direct consequence of the rotational invariance of the system. For a spherical symmetric system H commutes with L_z and $\mathbf{L}^2$ and the energy eigenkets are thus eigenkets for L_z and $\mathbf{L}^2$ also. Since the angular dependence is the same for all systems with spherical symmetry we can isolate this using the direction eigenket $|\hat{\mathbf{n}}\rangle$, and write

$$\langle \hat{\mathbf{n}}|l,m\rangle = Y_l^m(\vartheta,\varphi) = Y_l^m(\hat{\mathbf{n}}). \tag{3.89}$$

We can think of $Y_l^m(\vartheta,\varphi)$ as the amplitude of the state characterized by l,m found in the direction of $\hat{\mathbf{n}}$ specified by the angles ϑ,φ. From eigenvalue equations such as

$$L_z|l,m\rangle = m\hbar|l,m\rangle \tag{3.90}$$

we can infer the dependence on $Y_l^m(\vartheta,\varphi)$. Multiplying by $\langle\hat{\mathbf{n}}|$ we find

$$\langle \hat{\mathbf{n}}|L_z|l,m\rangle = -i\hbar\partial_\varphi\langle \hat{\mathbf{n}}|l,m\rangle = m\hbar\langle \hat{\mathbf{n}}|l,m\rangle, \tag{3.91}$$

which is

$$-i\hbar\partial\varphi Y_l^m(\vartheta,\varphi) = m\hbar Y_l^m(\vartheta,\varphi), \tag{3.92}$$

and thus the dependence on φ of $Y_l^m(\vartheta,\varphi)$ must be $e^{im\varphi}$. Similarly, using the eigenvalue equation

$$\mathbf{L}^2|l,m\rangle = l(l+1)\hbar^2|l,m\rangle \tag{3.93}$$

we have

$$\langle \hat{\mathbf{n}}|\mathbf{L}^2|l,m\rangle = l(l+1)\hbar^2\langle \hat{\mathbf{n}}|l,m\rangle = -\hbar^2\left[\frac{1}{\sin^2\vartheta}\partial_\varphi^2 + \frac{1}{\sin\vartheta}\partial_\vartheta(\sin\vartheta\partial_\vartheta)\right]\langle \hat{\mathbf{n}}|l,m\rangle, \tag{3.94}$$

which is equivalent to

$$\left[\frac{1}{\sin\vartheta}\partial_\vartheta(\sin_\vartheta\,\partial\vartheta) + \frac{1}{\sin^2\vartheta}\partial_\varphi^2 + l(l+1)\right]Y_l^m(\vartheta,\varphi) = 0. \tag{3.95}$$

The orthogonality relation

$$\langle l',m'|l,m\rangle = \delta_{ll'}\delta_{mm'} \tag{3.96}$$

is equivalent to

$$\int_0^{2\pi} d\varphi \int_{-1}^{1} d\cos\vartheta Y_{l'}^{m'*}(\vartheta,\varphi) Y_l^m(\vartheta,\varphi) = \delta_{ll'}\delta_{mm'}. \tag{3.97}$$

Consider the case $m = l$. We have the equation

$$L_+|l,l\rangle = 0, \tag{3.98}$$

which leads to

$$-i\hbar e^{i\varphi}[i\partial_\varphi - \cot\vartheta\partial_\varphi]\langle\hat{\mathbf{n}}|l,l\rangle = 0, \tag{3.99}$$

and this partial differential equation is satisfied by

$$\langle\hat{\mathbf{n}}|l,l\rangle = Y_l^l(\vartheta,\varphi) = c_l e^{il\varphi}\sin^l(\vartheta), \tag{3.100}$$

with c_l the renormalization constant

$$c_l = \left[\frac{(-1)^l}{2^l l!}\right]\sqrt{\frac{(2l+1)(2l)!}{4\pi}}. \tag{3.101}$$

From this we can obtain subsequent $Y_l^m(\vartheta,\varphi)$, which for $m \geq 0$ is

$$Y_l^m(\vartheta,\varphi) = \frac{(-1)^l}{2^l l!}\sqrt{\frac{(2l+1)(l+m)!}{4\pi(l-m)!}} e^{im\varphi}\frac{1}{\sin^m\vartheta}\frac{d^{l-m}}{d(\cos\vartheta)^{l-m}}(\sin\vartheta)^{2l}, \tag{3.102}$$

and we define

$$Y_l^{-m} = (-1)^m[Y_l^m(\vartheta,\varphi)]^*. \tag{3.103}$$

Note here that we cannot have l or m be half-integer valued since this would give us a multivalued function since we acquire a minus sign under a 2π rotation.

3.6.3 Spherical Harmonics as Rotation Matrices

We can find a general direction eigenket by applying a rotation $\mathscr{D}(R)$ to $|\hat{\mathbf{z}}\rangle$ such that

$$|\hat{\mathbf{n}}\rangle = \mathscr{D}(R)|\hat{\mathbf{z}}\rangle. \tag{3.104}$$

We can find this by two rotation, first about the y-axis and then about the z-axis, which in the language of Euler rotations is

$$\mathscr{D}(R) = \mathscr{D}(\alpha = \varphi, \beta = \vartheta, \gamma = 0). \tag{3.105}$$

Inserting the completeness relation in the rotation we find

$$|\hat{\mathbf{n}}\rangle = \sum_l\sum_m \mathscr{R} = |l,m\rangle\langle l,m|\hat{\mathbf{z}}\rangle. \tag{3.106}$$

We find that $|\hat{\mathbf{n}}\rangle$ expanded in $|l,m\rangle$ contains all possible l. We now multiply by $\langle l,m'|$ on the left and find

$$\langle l,m'|\hat{\mathbf{n}}\rangle = \sum_m \mathscr{D}_{mm'}^{(l)}(\varphi,\vartheta,0)\langle l,m|\hat{\mathbf{z}}\rangle, \tag{3.107}$$

and we see that $\langle l,m|\hat{\mathbf{z}}\rangle = Y_l^(m*)(0,\varphi)$ where φ is undetermined. Using this we find

$$Y_l^{m**}(\vartheta,\varphi) = \sqrt{\frac{2l+1}{4\pi}}\mathscr{D}_{m'0}^{(l)}(\varphi,\vartheta,0) \tag{3.108}$$

or

$$D_{m0}^{(l)}(\varphi,\vartheta,0) = \sqrt{\frac{4\pi}{2l+1}}Y_L^{m*}(\vartheta,\varphi) \tag{3.109}$$

3.8 Addition of Angular Momenta

3.8.1 Simple Examples of Angular-Momentum Addition

We have only studied either spin $\frac{1}{2}$ systems with other degrees of freedom ignored, or space systems with the internal degrees of freedom, such as spin ignored. Before going into the formal theory we will consider some familiar examples.

We consider the state that is a direct product state of the position and spin kets $|\mathbf{x}', \updownarrow\rangle = |\mathbf{x}'\rangle \otimes |\updownarrow\rangle$, where operators in the space spanned by the position kets commute by the operators in the space spanned by the spin-space. The rotation operator will still be $\exp(i\mathbf{J} \cdot \hat{\mathbf{n}}\varphi/\hbar)$, but angular momentum, the generator of rotation, will be

$$\mathbf{J} = \mathbf{L} + \mathbf{S} = \mathbf{L} \otimes 1 + 1 \otimes \mathbf{S}. \tag{3.110}$$

Since $[\mathbf{L}, \mathbf{S}] = 0$ we can write

$$\mathscr{D}(R) = \mathscr{D}^{(\text{orb})} \otimes \mathscr{D}^{(\text{spin})} = \exp\left(-\frac{i\mathbf{L} \cdot \hat{\mathbf{n}}}{\hbar}\varphi\right) \otimes \exp\left(-\frac{i\mathbf{S} \cdot \hat{\mathbf{n}}}{\hbar}\varphi\right). \tag{3.111}$$

We can write the wave function for a spin particle as

$$\langle \mathbf{x}', \updownarrow | \alpha \rangle = \psi_{\updownarrow}(\mathbf{x}') \tag{3.112}$$

or as a column matrix

$$\begin{pmatrix} \psi_{\uparrow}(\mathbf{x}') \\ \psi_{\downarrow}(\mathbf{x}') \end{pmatrix}. \tag{3.113}$$

We might change the representation of the space part to be the eigenket $|n, l, m\rangle$.

If we instead consider a spin $\frac{1}{2}$ two particle system, without considering the orbital angular momentum we can write the total spin operator

$$\mathbf{S} = \mathbf{S}_1 + \mathbf{S}_2 = \mathbf{S}_1 \otimes 1 + 1 \otimes \mathbf{S}_2, \tag{3.114}$$

where it is evident that operators living in the two different spaces commute. The usual commutation relations for the operators in the same space, and for the total operator applies, and we can write the eigenvalues as

$$\mathbf{S}^2 = (\mathbf{S}_1 + \mathbf{S}_2)^2 : s(s+1)\hbar^2 \tag{3.115}$$
$$S_z = S_{1z} + S_{2z} : m\hbar \tag{3.116}$$
$$S_{1z} : m_1\hbar \tag{3.117}$$
$$S_{2z} : m_2\hbar. \tag{3.118}$$

We can expand the ket corresponding of an arbitrary spin of the two particle system by either eigenkets to $\mathbf{S}^2$ and S_z or eigenkets to S_{1z} and S_{2z}:

1. Expanding in the $\{m_1, m_2\}$ basis gives the possible kets

$$|\uparrow\uparrow\rangle, |\uparrow\downarrow\rangle, |\downarrow\uparrow\rangle, \text{and } |\downarrow\downarrow\rangle. \tag{3.119}$$

2. Expanding in the $\{s, m\}$ basis (also called the triplet-singlet representation) gives the possible kets ($|s, m\rangle$)

$$|0, 0\rangle, |1, -1\rangle, |1, 0\rangle, \text{and } |1, 1\rangle. \tag{3.120}$$

The relationship between these two bases are

$$|0,0\rangle = \frac{1}{\sqrt{2}}(|\uparrow\downarrow\rangle - |\downarrow\uparrow\rangle) \tag{3.121}$$

$$|1,-1\rangle = |\downarrow\downarrow\rangle \tag{3.122}$$

$$|1,0\rangle = \frac{1}{\sqrt{2}}(|\uparrow\downarrow\rangle + |\downarrow\uparrow\rangle) \tag{3.123}$$

$$|1,1\rangle = |\uparrow\uparrow\rangle. \tag{3.124}$$

The coefficients that appear on the right-hand side are the simplest version of the Clebsch-Gordan coefficients. These coefficients are the matrix elements of the matrix that transforms between two different bases.

3.8.2 Formal Theory of Angular-Momentum Addition

We now consider two angular-momentum operators $\mathbf{J}_1$ and $\mathbf{J}_2$, living in different subspaces. The individual components of each operator satisfies the normal commutation relations, but they commute with the components of the other operator. Looking at the infinitesimal rotation of both subspaces simultaneously, we see

$$\left(1 - i\frac{\mathbf{J}_1 \cdot \hat{\mathbf{n}}}{\hbar}\delta\varphi\right) \otimes \left(1 - i\frac{\mathbf{J}_2 \cdot \hat{\mathbf{n}}}{\hbar}\delta\varphi\right) = 1 - i\frac{(\mathbf{J}_1 \otimes 1 + 1 \otimes \mathbf{J}_2) \cdot \hat{\mathbf{n}}}{\hbar}\delta\varphi. \tag{3.125}$$

We define the total angular momentum operator as

$$\mathbf{J} = \mathbf{J}_1 \otimes 1 + 1 \otimes \mathbf{J}_2 = \mathbf{J}_1 + \mathbf{J}_2. \tag{3.126}$$

For a finite rotation we have

$$\mathscr{D}_1(R) \otimes \mathscr{D}_2(R) = \exp\left(-i\frac{\mathbf{J}_1 \cdot \hat{\mathbf{n}}}{\hbar}\varphi\right) \otimes \exp\left(-i\frac{\mathbf{J}_2 \cdot \hat{\mathbf{n}}}{\hbar}\varphi\right). \tag{3.127}$$

Note the apparent similarities of the angle and axis of rotation for both subspaces. Importantly, the components of $\mathbf{J}$ satisfy the commutation relation for angular momentum. Physically the total angular momentum operator is the generator of rotation for the entire system. For choosing base kets there are two options:

1. The simultaneous eigenkets of $\mathbf{J}_1^2, \mathbf{J}_2^2, J_{1z}$, and J_{2z}, or the $\{j_1, j_2, m_1, m_2\}$ basis, denoted $|j_1, j_2; m_1, m_2\rangle$. The defining equations are

$$\mathbf{J}_1^2|j_1, j_2; m_1, m_2\rangle = j_1(j_1+1)\hbar^2|j_1, j_2; m_1, m_2\rangle \tag{3.128a}$$

$$\mathbf{J}_2^2|j_1, j_2; m_1, m_2\rangle = j_2(j_2+1)\hbar^2|j_1, j_2; m_1, m_2\rangle \tag{3.128b}$$

$$J_{1z}|j_1, j_2; m_1, m_2\rangle = m_1\hbar|j_1, j_2; m_1, m_2\rangle \tag{3.128c}$$

$$J_{2z}|j_1, j_2; m_1, m_2\rangle = m_2\hbar|j_1, j_2; m_1, m_2\rangle \tag{3.128d}$$

2. The simultaneous eigenkets of $\mathbf{J}_1^2, \mathbf{J}_2^2, \mathbf{J}^2$, and J_z or the $\{j_1, j_2; j, m\}$ basis. We note that this set commutes, denoted $|j_1, j_2; j, m\rangle$. The defining equation are

$$\mathbf{J}_1^2|j_1, j_2; j, m\rangle = j_1(j_1+1)\hbar^2|j_1, j_2; j, m\rangle \tag{3.129a}$$

$$\mathbf{J}_2^2|j_1, j_2; j, m\rangle = j_2(j_2+1)\hbar^2|j_1, j_2; j, m\rangle \tag{3.129b}$$

$$\mathbf{J}^2|j_1, j_2; j, m\rangle = j(j+1)\hbar^2|j_1, j_2; j, m\rangle \tag{3.129c}$$

$$J_z|j_1, j_2; j, m\rangle = m\hbar|j_1, j_2; j, m\rangle. \tag{3.129d}$$

Commonly the values for j_1, j_2 are implicit, and the ket is denoted as just $|j, m\rangle$.

It is important to note that

$$\left[\mathbf{J}^2, J_z\right] = 0, \quad \left[\mathbf{J}^2, J_{1z}\right] \neq 0, \quad \left[\mathbf{J}^2, J_{2z}\right] \neq 0. \tag{3.130}$$

We can therefore not mix these two representations. Consider the connecting transformation for the two bases

$$|j_1, j_2; j, m\rangle = \sum_{m_1} \sum_{m_2} |j_1, j_2; m_1, m_2\rangle \langle j_1, j_2; m_1, m_2 | j_1, j_2; j, m\rangle. \tag{3.131}$$

The elements $\langle j_1, j_2; m_1, m_2 | j_1, j_2; j, m\rangle$ are the Clebsch-Gordan coefficients.

We will now study some properties of the Clebsch-Gordan coefficients. If

$$m \neq m_1 + m_2, \tag{3.132}$$

the coefficients vanish. The coefficients also vanish unless

$$|j_1 - j_2| \leq j \leq j_1 + j_2. \tag{3.133}$$

The Clebsch-Gordan coefficients also form a unitary matrix, and we, by convention, take the elements to be real. A consequence of this (being a real unitary matrix) is that the matrix is orthogonal. We can write the Clebsch-Gordan coefficients in Wigner's 3-j symbol

$$\langle j_1, j_2; m_1, m_2 | j_1, j_2; j, m\rangle = (-1)^{j_1 - j_2 + m} \sqrt{2j+1} \begin{pmatrix} j_1 & j_2 & j \\ m_1 & m_2 & -m \end{pmatrix}. \tag{3.134}$$

3.8.3 Recursion Relations for the Clebsch-Gordan Coefficients

3.8.4 Clebsch-Gordan Coefficients and Rotation Matrices

3.11 Tensor Operators

3.11.1 Vector Operator

3.11.2 Cartesian Tensors Versus Irreducible Tensors

3.11.3 Product of Tensors

3.11.4 Matrix Elements of Tensor Operators; The Wigner-Eckart Theorem

4 Symmetry in Quantum Mechanics

6 Scattering Theory

Scattering processes are the processes that transform an initial state into a final state, by some potential, which we will treat as a time-dependent perturbation. This is how we learn about experimental processes.

6.1 Scattering as a Time-Dependent Perturbation

We make the assumption that the Hamiltonian may be written as

$$H = H_0 + V(\mathbf{x}), \tag{6.1}$$

with

$$H_0 = \frac{\mathbf{p}^2}{2m}, \tag{6.2}$$

the kinetic energy operator. The eigenvalues of H_0 are

$$E_{\mathbf{k}} = \frac{\hbar^2 \mathbf{k}^2}{2m}. \tag{6.3}$$

Plane wave eigenvectors of H_0, with eigenvalues $E_{\mathbf{k}}$, are $|\mathbf{k}\rangle$, and we make the assumption that the scattering potential $V(\mathbf{r})$ is time independent. We will treat this as an incoming particle seeing the scattering potential which is active only when the particle is close to the scatterer. This analysis will be performed using time-dependent perturbation theory in the interaction picture.

We know that the state evolution is

$$|\alpha, t_0; t\rangle_I = U_I(t, t_0)|\alpha, t_0; t_0\rangle, \tag{6.4}$$

where $U_I(t, t_0)$ is the time evolution operator in the interaction picture, satisfying

$$i\hbar\partial_t U_I(t, t_0) = V_I(t) U_I(t, t_0). \tag{6.5}$$

We have $U_I(t_0, t_0) = 1$ and $V_I(t) = \exp(iH_0 t/\hbar) V \exp(-iH_0 t/\hbar)$, and the exact solution to the equation may be written as

$$U_I(t, t_0) = 1 - \frac{i}{\hbar}\int_{t_0}^{t} dt' V_I(t') U_I(t', t_0). \tag{6.6}$$

The transition amplitude for the transformation from an initial state $|i\rangle$ into a final state $|n\rangle$, both eigenstate to H_0, can be written as

$$\langle n|U_I(t, t_0)|i\rangle = \delta_{ni} - \frac{i}{\hbar}\sum_m \langle n|V|m\rangle \int_{t_0}^{t} dt' e^{i\omega_{nm}t'} \langle m|U_I(t', t_0)|i\rangle, \tag{6.7}$$

where $\hbar\omega_{nm} = E_n - E_m$ and $\delta ni = \langle n|i\rangle$. We arrive here by inserting the completeness relation $\sum_m |m\rangle\langle m|$, and converting $V_I \to V$.

In scattering theory we deal with a continuum of states, not discrete ones, we therefore have to adjust the above equation to account for this. We do this by quantizing the scattering states in a cubic volume of side L. We have, in the coordinate representation, the wave function

$$\langle \mathbf{x}|\mathbf{k}\rangle = \frac{1}{L^{3/2}} e^{i\mathbf{k}\cdot\mathbf{x}}, \tag{6.8}$$

where we will take $L \to \infty$ at the end. Note that $\langle \mathbf{k'}|\mathbf{k}\rangle = \delta_{\mathbf{k'k}}$.

The initial and final states exist only asymptomatically, and we therefore have to have $t \to \infty$ and $t_0 \to -\infty$.

We start by doing a first order treatment, where we set $\langle m|U_I(t',t_0)|i\rangle = \delta_{mi}$, which gives

$$\langle n|U_I(t,t_0)|i\rangle = \delta_{ni} - \frac{i}{\hbar}\langle n|V|i\rangle \int_{t_0}^{t} dt' e^{i\omega_{ni}t'}. \tag{6.9}$$

For the $t \to \infty$ we have Fermi's golden rule, and in order to include the behaviour for $t_0 \to -\infty$ we define a matrix T_{ni} by

$$\langle n|U_I t, t_0|i\rangle = \delta_{ni} - \frac{i}{\hbar}T_{ni} \int_{t_0}^{t} dt' e^{i\omega_{ni}t'+\varepsilon t'} \tag{6.10}$$

with $\varepsilon > 0$ and $t \ll 1/\varepsilon$, and we need to make sure to take the limit $\varepsilon \to 0$ before we take $t \to \infty$. We now define the scattering matrix

$$S_{ni} = \lim_{t\to\infty}\left[\lim_{\varepsilon\to 0}\langle n|U_I t, -\infty|i\rangle\right] = \delta_{ni} - \frac{i}{\hbar}T_{ni}\int_{-\infty}^{\infty} dt' e^{i\omega nit'} = \delta_{ni} - 2\pi i\delta(E_n - E_i)T_{ni}. \tag{6.11}$$

We can see that the scattering matrix consists of two parts. One part where the final and initial states are the same, and one part governed by the T-matrix, with some sort of scattering.

6.1.1 Transition Rates and Cross Sections

We define that transition rate as

$$w(i \to n) = \partial_t |\langle n|U_I t, -\infty|i\rangle|^2 \tag{6.12}$$

and for $|i\rangle \neq |n\rangle$, we have

$$\langle n|U_I(t,-\infty)|i\rangle = -\frac{i}{\hbar}T_{ni}\int_{-\infty}^{t} dt' e^{i\omega_{ni}t'+\varepsilon t'} = -\frac{i}{\hbar}T_{ni}\frac{e^{i\omega_{ni}t+\varepsilon t}}{i\omega_{ni}+\varepsilon}. \tag{6.13}$$

The transition rate for this is then

$$w(i \to n) = \partial_t\left[\frac{1}{\hbar^2}|T_{ni}|^2\frac{e^{2\varepsilon t}}{\omega_{ni}^2+\varepsilon^2}\right] = \frac{1}{\hbar^2}|T_{ni}|^2\frac{2\varepsilon e^{2\varepsilon t}}{\omega_{ni}^2+\varepsilon^2}. \tag{6.14}$$

Here, we need to take $\varepsilon \to 0$ first, before taking $t \to \infty$. If $\omega_{ni} \neq 0$, we see that $w \to 0$, while for $\omega_{ni} = 0$, we get a delta function. We find that

$$\frac{ee^{2\varepsilon t}}{\omega_{ni}^2+\varepsilon^2} = \pi\delta(\omega_{ni}) = \pi\hbar\delta(E_n - E_i), \tag{6.15}$$

and we get the transmission rate

$$w(i \to n) = \frac{2\pi}{\hbar}|T_{ni}|^2\delta(E_n - E_i) \tag{6.16}$$

which is independent of time so the limit $t \to \infty$ is trivial. This is very similar to Fermi's golden rule, but with T_{ni} instead of V_{ni}, which is a more general matrix. We can use this transition rate to determine a scattering cross section.

Similarly to Fermi's golden rule, we need to determine the density of final states, $\rho(E_n) = \Delta n/\Delta E_n$, so that we can integrate over the final state energy. We will consider

elastic scattering, where $|i\rangle = |\mathbf{k}\rangle, |n\rangle = |\mathbf{k}'\rangle$, and $|\mathbf{k}| = |\mathbf{k}'| = k$. Using the normalization due to our cubic volume, we can write

$$E_{\mathbf{n}} = \frac{\hbar^2 \mathbf{k}'^2}{2m} = \frac{\hbar^2}{2m}\left(\frac{2\pi}{L}\right)^2 |\mathbf{n}|^2, \tag{6.17}$$

and thus

$$\Delta E_{\mathbf{n}} = \partial_{\mathbf{n}}(\Delta E_{\mathbf{n}})\Delta|\mathbf{n}| = \frac{\hbar^2}{m}\left(\frac{2\pi}{L}\right)^2 |\mathbf{n}|\Delta|\mathbf{n}|, \tag{6.18}$$

where $\mathbf{n} = n_x\hat{\mathbf{i}} + n_y\hat{\mathbf{j}} + n_z\hat{\mathbf{k}}$ and $n_{x,y,z} \in \mathbb{Z}$. We have $\mathbf{n} = (L/2\pi)|\mathbf{k}'| = (L/2\pi)k$ and L is large, so we can imagine $|\mathbf{n}|$ as almost continuous. The number of states within a spherical shell of radius $|\mathbf{n}|$ and thickness $\Delta|\mathbf{n}|$ is

$$\Delta n = 4\pi|\mathbf{n}|^2\Delta|\mathbf{n}| \times \frac{d\Omega}{4\pi}, \tag{6.19}$$

considering the fraction of solid angle represented by the final state vector $\mathbf{k}'$. Thus,

$$\rho(E_n) = \frac{\Delta n}{\Delta E_n} = \frac{m}{\hbar^2}\left(\frac{L}{2\pi}\right)^2 |\mathbf{n}|d\Omega = \frac{mk}{\hbar^2}\left(\frac{L}{2\pi}\right)^3 d\Omega. \tag{6.20}$$

Integrating over the final states gives

$$w(i \to n) = \frac{mkL^3}{(2\pi)^2\hbar^3}|T_{ni}|^2 d\Omega. \tag{6.21}$$

This concept of cross section is used to understand the transition rate in scattering experiments. Determining the rate at which particles scatter into a solid angle $d\Omega$ from a beam of particles with momentum $\hbar\mathbf{k}$. The speed of the particles is $v = \hbar k/m$, and the time needed to cross the cubic volume is $t = L/v$. The particle flux in the beam is v/L^3, and the probability flux for the wave function becomes

$$\mathbf{j}(\mathbf{x}, t) = \left(\frac{\hbar}{m}\right)\frac{\mathbf{k}}{L^3} = \frac{v}{L^3}. \tag{6.22}$$

The cross section $d\sigma$ is defined as the transition rate divided by the flux. Finally, we arrive at

$$\frac{d\sigma}{d\Omega} = \left(\frac{mL^3}{2\pi\hbar^2}\right)^2 |T_{ni}|^2. \tag{6.23}$$

We now have to relate the matrix elements T_{ni} to the scattering potential $V(\mathbf{r})$.

6.1.2 Solving for the T matrix

Recalling the definition of the T matrix we have

$$\langle n|U_I(t, -\infty)|i\rangle = \delta_{ni} + \frac{1}{\hbar}T_{ni}\frac{e^{i\omega_{ni}t+\varepsilon t}}{-\omega_{ni} + i\varepsilon} \tag{6.24}$$

and

$$\langle n|U_I(t, -\infty)|i\rangle = \delta_{ni} - \frac{i}{\hbar}\sum_m V_{nm}\int_{-\infty}^{t} dt' e^{i\omega_{nm}t'}\langle m|U_I(t', -\infty)|i\rangle. \tag{6.25}$$

Inserting the former equation into the latter under the integral sign we obtain three terms. The first of which is δ_{ni}, the second looks like the former equation but with T_{ni} replaced by V_{ni}. The final term is

$$-\frac{i}{\hbar}\frac{1}{\hbar}\sum_m V_{nm}\frac{T_{mi}}{-\omega_{mi}+i\varepsilon}\int_{-\infty}^{t} dt' e^{i(\omega_{nm}+\omega_{mi})t'+\varepsilon t'}. \tag{6.26}$$

Since $\omega_{nm}+\omega_{mi}=\omega_{ni}$, the result of the integral can be taken out from the summation. Gathering terms and comparing to the former equation we find that

$$T_{ni} = V_{ni} + \frac{1}{\hbar}\sum_m V_{nm}\frac{T_{mi}}{-\omega_{mi}+i\varepsilon} = V_{ni} + \sum_m V_{nm}\frac{T_{mi}}{E_i - E_m + i\hbar\varepsilon}. \tag{6.27}$$

This is an inhomogeneous system of linear equation which can be solved for T_{ni} in terms of the matrix elements V_{nm}, which are known.

We can define a set of vectors $\left|\psi^{(+)}\right\rangle$ in terms of components in some basis $|j\rangle$, such that

$$T_{ni} = \sum_j \langle n|V|j\rangle\left\langle j\middle|\psi^{(+)}\right\rangle = \left\langle n\middle|V\middle|\psi^{(+)}\right\rangle. \tag{6.28}$$

Inserting this into the equation for T_{ni}, we find

$$\left\langle n\middle|V\middle|\psi^{(+)}\right\rangle = \langle n|V|i\rangle + \sum_m \langle n|V|m\rangle\frac{\left\langle m\middle|V\middle|\psi^{(+)}\right\rangle}{E_i - E_m + i\hbar\varepsilon}. \tag{6.29}$$

Since this must be true for all $|n\rangle$ we find

$$|\psi(+)\rangle = |i\rangle + \frac{1}{E_i - H_0 + i\hbar\varepsilon}V\left|psi^{(+)}\right\rangle, \tag{6.30}$$

which is known as the Lippmann-Schwinger equation. We can define an operator T with matrix elements T_{ni} by writing

$$T|i\rangle = V\left|\psi^{(+)}\right\rangle, \tag{6.31}$$

we can then find the operator equation

$$T = V + V\frac{1}{E_i - H_0 + i\hbar\varepsilon}T. \tag{6.32}$$

6.1.3 Scattering from the Future to the Past

We could instead imagine the scattering process evolving backwards in time, from a plane wave state $|i\rangle$ in the far future, to a state $|n\rangle$ in the distant past. We would in this case write

$$U_I(t,t_0) = 1 + \frac{i}{\hbar}\int_t^{t_0} dt' V_I(t')U_I(t',t_0) \tag{6.33}$$

and the T matrix is the n defined as

$$\langle n|U_I(t,t_0)|i\rangle = \delta_{ni} + \frac{1}{\hbar}T_{ni}\int_t^{t_0} dt' e^{i\omega_{ni}t'-\varepsilon t'}. \tag{6.34}$$

In this case we can define T using a different set of states $\left|\psi^{(-)}\right\rangle$ through

$$T|i\rangle = V\left|\psi^{(-)}\right\rangle. \tag{6.35}$$

7 Identical Particles

8 Relativistic Quantum Mechanics